

Exercise Class 1 — Consumer Theory (MWG Ch. 1–3)

Advanced Microeconomics 1 · Thursday 8 October, 11:00–13:00

Format (120 min). 60 min: assigned students present Problems 1–5 at the board (assignments posted Monday 5 October). 30 min: instructor works the hard item (Problem 6). 30 min: timed prelim-style question (Problem 7), closed book, ungraded, self-marked against the sketch afterwards.

Part I — Board problems

Problem 1 Rational preferences and their limits

- (a) Let $\succsim$ be rational. Prove that $\succ$ is transitive and irreflexive, and that $\sim$ is an equivalence relation.
- (b) Give an example of a complete but intransitive relation on a three-element set, and exhibit an explicit money pump against an agent with those strict preferences.
- (c) Lexicographic preferences on $\mathbb{R}_+^2$: $x \succsim y$ iff $[x_1 > y_1]$ or $[x_1 = y_1 \text{ and } x_2 \geq y_2]$. Show they are rational and strongly monotone; show they are not continuous; sketch why no utility representation exists (map each x_1 to a rational number in a suitably chosen interval and count).
- (d) For finite X and rational $\succsim$, verify that $u(x) = \#\{y \in X : x \succsim y\}$ represents $\succsim$.

Problem 2 Choice structures and WARP on a finite domain

Let $X = \{a, b, c\}$ and $\mathcal{B} = \{\{a, b\}, \{b, c\}, \{a, c\}\}$ (pairs only), with

$$C(\{a, b\}) = \{a\}, \quad C(\{b, c\}) = \{b\}, \quad C(\{a, c\}) = \{c\}.$$

- (a) Verify that this choice structure satisfies WARP.
- (b) Show that no rational preference rationalizes it.
- (c) Now add $\{a, b, c\}$ to $\mathcal{B}$. Show that *no* specification of $C(\{a, b, c\})$ preserves WARP. Reconcile with MWG Proposition 1.D.2.

Problem 3 The CES workout

Let $u(x_1, x_2) = (x_1^\rho + x_2^\rho)^{1/\rho}$, $\rho < 1$, $\rho \neq 0$; write $\sigma = 1/(1 - \rho)$ and $P = (p_1^{1-\sigma} + p_2^{1-\sigma})^{1/(1-\sigma)}$.

- (a) Derive $x(p, w)$ and $v(p, w)$ from the UMP.

- (b) Derive $h(p, u)$ and $e(p, u)$ from the EMP.
- (c) Verify all four duality identities: $e(p, v(p, w)) = w$; $v(p, e(p, u)) = u$; $h(p, u) = x(p, e(p, u))$; $x(p, w) = h(p, v(p, w))$.
- (d) Verify Roy's identity and Shephard's lemma by direct differentiation.
- (e) Recover Cobb–Douglas as $\sigma \rightarrow 1$ and Leontief as $\sigma \rightarrow 0$ (limits of $e(p, u)$ suffice).
- (f) *The two accounting laws at work.* For a general demand system satisfying homogeneity of degree zero and Walras' law, differentiate the two identities to derive **Engel aggregation** $\sum_{\ell} p_{\ell} \partial x_{\ell} / \partial w = 1$ and **Cournot aggregation** $\sum_{\ell} p_{\ell} \partial x_{\ell} / \partial p_k = -x_k$. Conclude that not every good can be inferior. Then verify both identities explicitly for the CES system of part (a).

Problem 4 Quasilinear preferences: the welfare-exact case

Let $u(x_0, x_1) = x_0 + \phi(x_1)$ with $\phi' > 0 > \phi''$, $p_0 = 1$, and w large enough for interior solutions.

- (a) Derive $x_1(p_1)$ (note: independent of w), $v(p_1, w)$, $e(p_1, u)$, $h_1(p_1)$.
- (b) Show $h_1(p_1, u) = x_1(p_1)$ for every u : Hicksian and Walrasian demand coincide.
- (c) For a price change $p_1^0 \rightarrow p_1^1$, prove $CV = EV = \Delta CS = \int_{p_1^1}^{p_1^0} x_1(p) dp$.
- (d) Exactly which step fails when wealth effects are restored? Point to the term in the Slutsky equation.

Problem 5 Slutsky by hand, once in your life

For the CES demand system of Problem 3 with $\sigma = 2$, $p = (1, 1)$, $w = 2$:

- (a) Compute the 2×2 Slutsky matrix $S(p, w) = D_p x + D_w x x^{\top}$ numerically.
- (b) Verify symmetry, negative semidefiniteness (compute eigenvalues or check $s_{11} \leq 0$ and $\det S = 0$), and $S p = 0$.
- (c) Confirm $S = D_p h$ evaluated at $u = v(p, w)$.
- (d) With $L = 2$, show from NSD + singularity alone that $s_{12} \geq 0$: two goods must be net substitutes.

Part II — Instructor-worked hard item

Problem 6 Afriat by hand

Consider $T = 3$ observations on two goods:

t	p_1^t	p_2^t	x_1^t	x_2^t
1	2	1	3	2
2	1	2	2	3
3	1	1	4	1

- (a) Build the 3×3 expenditure cross-matrix $E_{ts} = p^t \cdot x^s$ and the direct revealed preference relations R^D (chosen bundle weakly cheaper comparison) and P^D .
- (b) Compute the transitive closure and check GARP.
- (c) If GARP holds, find Afriat numbers (U_t, λ_t) solving $U_s \leq U_t + \lambda_t p^t(x^s - x^t)$ for all s, t , and write down the rationalizing utility $u(x) = \min_t \{U_t + \lambda_t p^t(x - x^t)\}$. If it fails, exhibit the violating cycle(s) and compute the CCEI by hand (find the budget scaling at which the last cycle just breaks).
- (d) Now replace the third observation by $x^3 = (2, 2)$ (same prices). Re-check GARP; construct (U_t, λ_t) — try $\lambda_t = 1$ — and the rationalizing $u(x) = \min_t \{U_t + \lambda_t p^t(x - x^t)\}$; evaluate u at the three bundles and confirm each is optimal in its own budget set.

Part III — Timed prelim question (30 minutes, closed book)

Problem 7 Reconstructing the square from v

A consumer's indirect utility is

$$v(p_1, p_2, w) = \frac{w}{p_1^\alpha p_2^{1-\alpha}}, \quad \alpha \in (0, 1).$$

- (a) Verify v has the properties required of an indirect utility function (h.d.0, monotonicity, quasi-convexity may be asserted with a one-line reason).
- (b) Recover Walrasian demand via Roy's identity.
- (c) Recover $e(p, u)$ by inverting, and Hicksian demand via Shephard's lemma.
- (d) Compute the Slutsky matrix and verify symmetry, NSD, and $Sp = 0$.
- (e) Which familiar preferences generated this v ? Justify.

Part IV — Take-home extension

Problem 8 Cheapflation and the limits of the superlative index

After Chen, Levell & O’Connell (2025), IFS WP 25/21, from Lecture 4. Not presented at the board; a full worked solution is in the instructor annex — self-mark, then bring questions to office hours.

One physical good — bread — comes in two quality rungs: cheap (good 1) and premium (good 2). A household has Stone–Geary preferences

$$u(q_1, q_2) = (q_1 - \gamma)^\alpha q_2^{1-\alpha}, \quad \alpha = \frac{1}{2}, \quad \gamma = 1,$$

read as: one loaf of cheap bread per period is committed subsistence; taste is Cobb–Douglas above it. Initial prices $p^0 = (1, 2)$; nominal spending is *fixed* at $x = 4$ in both periods (stagnant incomes). The surge is cheapflation: $p^1 = (1.35, 2.30)$ — the cheap rung +35%, the premium rung +15%.

- (a) For general $\alpha \in (0, 1)$, $\gamma \geq 0$: derive Walrasian demand, the indirect utility function, and show

$$e(p, u) = \gamma p_1 + u \left(\frac{p_1}{\alpha} \right)^\alpha \left(\frac{p_2}{1 - \alpha} \right)^{1-\alpha}.$$

For which parameter value are these preferences homothetic?

- (b) Compute the period-0 bundle, u^0 , and the budget share of cheap bread.
- (c) Compute the Laspeyres index P_L and the Konüs index at initial living standards, $P_K(u^0)$. Verify $P_L \geq P_K(u^0)$ and interpret the gap.
- (d) Compute the period-1 bundle and u^1 . Show the cheap-bread budget share *rises* even though the cheap rung’s relative price rose. Decompose the share change using the compensated (Hicksian) share at (p^1, u^0) : how much is substitution, how much is trading down (the income effect)?
- (e) Compute the Paasche index P_P , the Fisher index $P_F = \sqrt{P_L P_P}$, and $P_K(u^1)$. Establish the full ordering of all five numbers. Which of the classical index-number inequalities survive non-homotheticity, and why does the “superlative” index approximate *neither* true cost-of-living index here?
- (f) Now a richer household: same preferences, $x = 10$ in both periods. Compute its $P_K(u^0)$ and the poor–rich gap in true inflation. Relate this to Chen–Levell–O’Connell’s 7.4pp finding, and state — with a defence in terms of CV or EV — which of part (e)’s five numbers automatic indexation *should* pay this household.

Solutions and board plan

Board plan. P1 (a,c) one student, (b,d) another — 12 min total. P2 one student — 10 min; the punchline to draw out: pairs-only domains never test transitivity, which is exactly the message of 1.D.2. P3 split (a,b) / (c,d,e) / (f) across three students — 24 min; enforce the price-index notation, it pays for itself all course. P4 one student — 10 min. P5 one student with a laptop or fast hands — 8 min.

P1 sketches. (b) $a \succ b \succ c \succ a$; pump: start holding c , sell c buy b paying ε , then $b \rightarrow a$, then $a \rightarrow c$, repeat — unbounded extraction. (c) Continuity fails on $x^n = (1/n, 0) \rightarrow (0, 0)$ vs $y = (0, 1)$: $x^n \succ y$ for all n but $y \succ \lim x^n$. Representation: if u existed, for each x_1 pick a rational $q(x_1) \in (u(x_1, 0), u(x_1, 1))$; monotonicity in the first coordinate makes q injective from $\mathbb{R}$ into $\mathbb{Q}$ — impossible.

P2 sketches. (a) WARP never bites: no two budget sets share two alternatives. (b) Rationalization would force $a \succ b \succ c \succ a$ (each singleton choice from a pair reveals strict preference under rationalization), contradicting transitivity. (c) Whatever is chosen from the triple, some pair {chosen, other} contradicts the pairwise data via WARP. E.g. $a \in C(\{a, b, c\})$: take $B' = \{a, c\}$ where c was chosen with a available and $a \notin C(B')$ — violation. Consistent with 1.D.2, which *requires* the triple to be in $\mathcal{B}$ for its conclusion; here adding the triple makes WARP itself unsatisfiable, revealing the latent intransitivity.

P3 answers to check against.

$$x_\ell = \frac{p_\ell^{-\sigma} w}{P^{1-\sigma}}, \quad v = \frac{w}{P}, \quad e = u P, \quad h_\ell = u p_\ell^{-\sigma} P^\sigma.$$

Common slip in (d): forgetting $\partial P / \partial p_\ell = (p_\ell / P)^{-\sigma}$. For (f): differentiate $x(\alpha p, \alpha w) = x(p, w)$ in α at $\alpha = 1$ to get $\sum_k p_k \partial x_\ell / \partial p_k + w \partial x_\ell / \partial w = 0$ (Euler); differentiate $p \cdot x = w$ in w (Engel) and in p_k (Cournot). Inferior-goods claim: if all $\partial x_\ell / \partial w < 0$ the Engel sum is negative, not 1. CES check: $x_\ell = w p_\ell^{-\sigma} P^{\sigma-1}$; the sums telescope via $\sum_\ell p_\ell^{1-\sigma} = P^{1-\sigma}$.

P4 sketches. x_1 solves $\phi'(x_1) = p_1$, so $x_1(p_1) = (\phi')^{-1}(p_1)$; $v = w - p_1 x_1(p_1) + \phi(x_1(p_1))$; $e(p_1, u) = u - [\phi(x_1(p_1)) - p_1 x_1(p_1)]$. Since $\partial x_1 / \partial w = 0$, the Slutsky income term is zero, the three demand curves coincide, and the three welfare measures are the same integral. (d) The failing step is $h_1(p, u)$ depending on u once $\partial x_1 / \partial w \neq 0$.

P5 numbers. At $\sigma = 2$, $p = (1, 1)$, $w = 2$: $x = (1, 1)$; $\partial x_1 / \partial p_1 = -3/2$, $\partial x_1 / \partial p_2 = 1/2$, $\partial x_1 / \partial w = 1/2$ (and symmetrically). $S = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}$: symmetric; eigenvalues $\{0, -2\}$ (NSD); $S(1, 1)^\top = 0$. (*Re-derive before class; the point of the exercise is the checking ritual, not these particular numbers.*)

P6 worked notes. Expenditure cross-matrix $E_{ts} = p^t \cdot x^s$: rows $(8, 7, 9)$, $(7, 8, 6)$, $(5, 5, 5)$. At $e = 1$: $x^1 P^D x^2$ and $x^2 P^D x^1$ ($7 < 8$ both ways) — a strict two-cycle, so GARP fails outright; additionally $x^2 P^D x^3$ ($6 < 8$) while $x^3 R^D x^2$ ($5 = 5$), a second violation. CCEI: every relation *out of* x^3 requires $5e \geq 5$, so all cycles through x^3 die for any $e < 1$; the binding cycle is $x^1 - x^2$, alive iff $8e \geq 7$ with a strict arrow iff $8e > 7$. GARP therefore holds exactly on $e \leq 7/8$: **CCEI** $= 7/8 = 0.875$. For part (d), the consistent dataset (same prices) $x^1 = (2, 4)$, $x^2 = (4, 2)$, $x^3 = (2, 2)$ has expenditures $(8, 8, 4)$

and cross-matrix rows (8, 10, 6), (10, 8, 6), (6, 6, 4): the only relations are $x^1 P^D x^3$ and $x^2 P^D x^3$ — no cycles, GARP holds. With $\lambda_t = 1$, the inequalities pin $U_3 = U_1 - 2 = U_2 - 2$ (the $s=3$ constraints bind against the $t=3$ ones); take $(U_1, U_2, U_3) = (2, 2, 0)$, giving

$$u(x) = \min\{2x_1 + x_2 - 6, \quad x_1 + 2x_2 - 6, \quad x_1 + x_2 - 4\}.$$

Check: $u(x^1) = u(x^2) = 2$, $u(x^3) = 0$ (each equals its U_t , with the own affine piece binding), and on each budget $\{x : p^t x \leq p^t x^t\}$ the own piece bounds u by U_t , attained at x^t — optimality confirmed. *These are the practical's toy dataset: the class has already computed, in R, that the first triple violates GARP three times with CCEI = 7/8. Today they prove the same facts by hand — close that loop for them explicitly.*

P7 sketches. (b) Roy: $x_1 = -\frac{\partial v/\partial p_1}{\partial v/\partial w} = \alpha w/p_1$; similarly x_2 . (c) Set $v(p, e) = u$: $e(p, u) = u p_1^\alpha p_2^{1-\alpha}$; Shephard: $h_1 = \alpha u (p_2/p_1)^{1-\alpha}$, $h_2 = (1 - \alpha) u (p_1/p_2)^\alpha$. (d) $s_{11} = -\alpha(1 - \alpha)w/p_1^2 \leq 0$, $s_{12} = s_{21} = \alpha(1 - \alpha)w/(p_1 p_2)$, $s_{22} = -\alpha(1 - \alpha)w/p_2^2$; $\det S = 0$; $S p = 0$. (e) Cobb–Douglas $u = x_1^\alpha x_2^{1-\alpha}$ (up to monotone transformation) — constant expenditure shares are the giveaway.

P8 worked notes (take-home; all numbers verified). (a) Supernumerary income $S = x - \gamma p_1$; demands $q_1 = \gamma + \alpha S/p_1$, $q_2 = (1 - \alpha)S/p_2$; $v(p, x) = S/A(p)$ with $A(p) = (p_1/\alpha)^\alpha (p_2/(1 - \alpha))^{1-\alpha}$; inverting, $e(p, u) = \gamma p_1 + u A(p)$. Homothetic iff $\gamma = 0$ (plain Cobb–Douglas). (b) $S^0 = 3$: $q^0 = (2.5, 0.75)$, $u^0 = 3/(2\sqrt{2}) \approx 1.0607$, cheap share $s_1^0 = 62.5\%$. (c) $A(p^1) = 2\sqrt{1.35 \times 2.30} \approx 3.5242$; $e(p^1, u^0) \approx 5.0880$, so $P_K(u^0) = 1.2720$; $p^1 \cdot q^0 = 5.100$, so $P_L = 1.2750 \geq P_K(u^0)$ — the 0.30pp gap is the substitution effect, the $\frac{1}{2}\Delta p^\top S \Delta p$ of L3 made exact. (d) $S^1 = 2.65$: $q^1 \approx (1.9815, 0.5761)$, $u^1 \approx 0.7519 < u^0$; $s_1^1 = 66.9\%$. Compensated share at (p^1, u^0) : $h_1 = \gamma + \alpha(e(p^1, u^0) - \gamma p_1^1)/p_1^1 \approx 2.3844$, share $\approx 63.3\%$. Decomposition: +0.8pp substitution (the committed γ makes even the compensated share drift up), +3.6pp trading down — the income effect dominates, as in CLO. (e) $p^0 \cdot q^1 \approx 3.1337$, $P_P = 1.2765$; $P_F = 1.2757$; $e(p^0, u^1) \approx 3.1268$, $P_K(u^1) = 1.2793$. Ordering:

$$P_K(u^0) = 1.2720 < P_L = 1.2750 < P_F = 1.2757 < P_P = 1.2765 < P_K(u^1) = 1.2793.$$

What survives: $P_L \geq P_K(u^0)$ and $P_P \leq P_K(u^1)$ — each is a pure revealed-preference/feasibility bound at its *own* utility level, needing no homotheticity. What dies: $P_P \leq P_L$ and “Fisher $\approx$ the true index,” both of which need P_K to be independent of u . Here Fisher sits *between two different truths*, approximating neither — and lands *farther* from the CV-relevant $P_K(u^0)$ than Laspeyres does (0.37pp vs 0.30pp): trading down loads the final-period basket onto the fast-inflating rung, and the superlative reads that income effect as if it were substitution. This is CLO’s non-homotheticity bias in two goods. (f) Rich household: $u_r^0 = 9/(2\sqrt{2}) \approx 3.1820$, $P_K(u_r^0) = 1.2564$ — true inflation 25.6% vs the poor household’s 27.2%: a 1.6pp gap from one preference and one γ , the CLO/Jaravel gradient in miniature (their 7.4pp adds many rungs, many categories, and stronger cheapflation). Indexation defence: a CV-standard scheme owes $P_K(u^0)$ — 27.2% for the poor household, 25.6% for the rich; *any* single number over- or under-pays someone, and even the household-specific Fisher over-pays the CV standard here. An EV defence ($P_K(u^1)$, 27.9%) is coherent but must own that it compensates the *new, lower* living standard’s marginal basket. The point to reward: recognizing the choice is normative, and that “use a superlative index” does not resolve it.

Timing note. If Part I overruns, sacrifice P5 (the practical already computed it numerically) — never P6. P8 is take-home by design; if the room is strong, trail it in the last two minutes: “the Fisher index is usually the careful choice — Problem 8 is a world where it is careful and still wrong.”